Superstore Sales and Profitability

- ## Problem Statement

The dataset indicates that while overall sales volume is high, profitability is not consistent across product categories and sub-categories. Certain sub-categories generate substantial revenue but incur losses. Additionally, discounting patterns appear to negatively affect profit across multiple cases.

Objective:

- Analyze sales and profitability performance
- Identify loss-making areas
- Evaluate the impact of discounting on profit
- Provide actionable business recommendations

- ## Data Description

- Approx. 9,900 retail transactions
- Source: Superstore dataset

Variables:

- Category, Sub-Category
- Region
- Segment
- Sales, Profit, Discount, Quantity

- ## Data Preparation

Data preparation was performed using Python **(Pandas):**

- Duplicate records were identified and removed
- Dataset was checked for missing values **(no null values found)**
- Column names were standardized (lowercase, snake_case format)
- Data was prepared for aggregation and analysis

- ## Analytical Approach

- The analysis is based on aggregation and comparison of sales and profit across:
- Category and Sub-Category
- Region
- Customer Segment

Discount values were analyzed to understand their impact on profitability.

- ## Analysis and Findings

5.1 Sales Performance

Category-wise Sales

- Technology : **$836,154.03** (Highest)
- Office Supplies : **$429,292.68** (Lowest)

Segment-wise Sales

- Consumer : **$1,160,832.77** (Highest)
- Home Office : **$429,292.68** (Lowest)

Region-wise Sales

- West region recorded the highest sales
- South region recorded the lowest sales

5.2 Profitability

Category-wise Profit

- Technology : **$145,454.95** (Highest profit)
- Furniture: Low profitability despite strong sales

High sales do not necessarily translate into high profit

5.3 Sub-Category Performance

Top Profit-Generating sub-categories:

- Copiers
- Phones
- Accessories

Loss-making sub-categories:

- Tables : **-$17,725.48** (highest loss)
- Bookcases
- Supplies

Tables show high sales but negative profit.

5.4 Discount Impact

- Higher discounts are associated with lower profit
- Heavy discounting leads to loss-making transactions
- This effect is most significant in the Furniture category

5.5 Regional Performance

- West region : Highest profitability
- Central region : Lowest profitability

- ## Dashboard Summary

A Power BI dashboard was developed to present key performance indicators and trends.

Key Metrics:

- Total Sales: $2.3M
- Total Orders: 9,977
- Average Order Value: $230.15

Visuals Included:

- Sales by Category
- Sales by Region (percentage contribution)
- Sales by Segment
- Profit by Sub-Category
- Loss by Sub-Category

- **Figure : Power BI Dashboard Overview**

The dashboard provides a consolidated view of performance across categories, regions, and segments.

- **Key Insights**

- High sales do not guarantee profitability
- Tables sub-category causes major losses despite strong demand
- Discounting is the primary driver of losses
- Consumer segment contributes the highest revenue
- West region performs best overall

- ## Recommendations

Discount Control

- Reduce excessive discounting, particularly in the Furniture category and Tables sub-category.

Sub-Category Review

- Reassess pricing and cost structure for Tables, Bookcases, and Supplies.

Focus on Profitable Products

- Maintain emphasis on Copiers, Phones, and Accessories.

Regional Review

- Investigate factors contributing to lower profitability in the Central region.

Performance Monitoring

- Track sub-category performance to identify and address loss trends.

- ## Conclusion

The analysis shows that although the business generates strong sales, profitability is negatively affected by excessive discounting and underperforming sub-categories. Losses are concentrated mainly in the Furniture category, particularly in Tables. By optimizing discount strategies and improving pricing decisions, the business can significantly enhance profitability without increasing sales volume.

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